Counting convex subsets in iterated order-type blow-ups

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Abstract

For a planar point set P in general position, let $v(P)$ be the number of its subsets in convex position, and let

$$f(N) = \min\{v(P) : |P| = N\}.$$

Erdős and Hammer asked for the asymptotic behaviour of $f(N)$; a standard counting estimate applied to the classical construction gives $\log_2 f(N) \leq (1 + o(1))(\log_2 N)^2$. We prove

$$\limsup_{N\to\infty} \frac{\log_2 f(N)}{(\log_2 N)^2} \leq \frac{1}{2}.$$

The construction is a directionally specified iterated order-type blow-up. We give exact substitution formulas for the numbers of caps, cups, and convex subsets. If a fixed r-point template has largest cap and cup of sizes a and b, respectively, its iterates have normalized coefficient $(a + b - 2)/(2 \log_2 r)$. Balanced instances of the classical cup–cap construction make this coefficient tend to $1/2$. The cup–cap theorem shows that $1/2$ is optimal among fixed-template iterations. More strongly, we prove that every mirror-decomposable point set has logarithmic convex-subset count at least $(\log_2 N)^2/2 - O((\log N)^{3/2})$. Thus the optimal universal coefficient $1/2$ is attained within the entire decomposable class.

1 Introduction

For a finite set $P \subset \mathbb{R}^2$ in general position, call a subset of P *convex* if all of its points are vertices of its convex hull. Subsets of size at most two are included; changing this convention, or including the empty set, has no effect on any normalized asymptotic below. Write $v(P)$ for the number of convex subsets of P. Throughout the paper, all logarithms are base two.

Erdős [Erd78] recorded a question posed with Hammer: determine or estimate

$$f(N) = \min_{|P|=N} v(P),$$

where the minimum is over planar point sets in general position. The classical Erdős–Szekeres construction and the convex-polygon theorem show that $f(N) = 2^{\Theta((\log N)^2)}$; see, for example, the survey of Morris and Soltan [MS00]. The open problem asks in particular whether $\log f(N)/(\log N)^2$ has a limit.

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For completeness, the standard coefficient-1 upper estimate is immediate. Take $k = \lceil \log N \rceil + 2$ and retain N points from the classical 2^{k-2}-point construction with no convex k-set. Then

$$v(P) \le \sum_{j<k} \binom{N}{j} \le kN^{k-1} = 2^{(1+o(1))(\log N)^2}.$$

For context, Suk's bound $ES(k) = 2^{k+o(k)}$ [Suk17] gives the following standard lower estimate. Put $t = \lfloor N^{1/2} \rfloor$ and $k = (1-o(1)) \log t$, so every t-point set contains a convex k-set. Double-counting pairs (K, T) with $K \subseteq T \subseteq P$, $|K| = k$, $|T| = t$, gives

$$\#\{\text{convex } k\text{-subsets of } P\} \ge \frac{\binom{N}{t}}{\binom{N-k}{t-k}} = \frac{\binom{N}{k}}{\binom{t}{k}}.$$

Consequently

$$\liminf_{N\to\infty} \frac{\log f(N)}{(\log N)^2} \ge \frac{1}{4}.$$

Together with Theorem 1.1, this gives the rigorous window

$$\frac{1}{4} \le \liminf \frac{\log f(N)}{(\log N)^2} \le \limsup \frac{\log f(N)}{(\log N)^2} \le \frac{1}{2}.$$

Our result improves the coefficient in the classical upper bound.

Theorem 1.1. *One has*

$$\limsup_{N\to\infty} \frac{\log f(N)}{(\log N)^2} \le \frac{1}{2}.$$

The construction is a special realization of an order-type blow-up. General blow-ups and their iteration appear in Han–Kohayakawa–Sales–Stagni [HKSS19]; almost-vertical blow-ups tailored to the Erdős–Szekeres problem appear in Baek–Balko [BB26]. Recursively separated point sets were studied under the name *decomposable* by Balko, Kynčl, Langerman, and Pilz [BKLP17]. Baek and Balko proved the Erdős–Szekeres conjecture for that class [BB26, Theorem 7]. Our final result is enumerative: in the mirrored convention we determine the asymptotic minimum total number of convex subsets in the same class.

Exact weighted identities for numbers of convex polygons are also known [HOPTV22]. The feature used here is a prescribed orientation for triples meeting exactly two clusters. Those orientations give exact substitution identities for the three statistics in Lemma 2.2; iterating them is what produces the geometric coefficient $1/2$. The same base-normalized coefficient occurs in Székely's graph-theoretic analogue [Sz84], but that argument does not directly yield the geometric statement here.

We also determine the asymptotic answer throughout the class of mirror-decomposable point sets; see Theorem 5.1. The lower bound there uses a max-endpoint recurrence and a multiscale reset argument. It does not extend to arbitrary order types, so the existence and value of the limit in the Erdős–Hammer problem remain open.

2 A directional vertical composition

After a rotation, order every point set by increasing x-coordinate. For three points $p < q < r$, let $\chi(p, q, r) \in \{-, +\}$ denote their orientation. A *cap* is a nonempty set all of whose increasing triples have sign $-$, and a *cup* is defined with sign $+$. Singletons and pairs are both caps and cups.

A suitable shear $(x, y) \mapsto (x, y+Mx)$ preserves orientations and makes the y-coordinates increase in the same order as the x-coordinates. We shall therefore assume both coordinates increase.

Let $S = (s_1, \ldots, s_r)$ and $Q = (q_1, \ldots, q_n)$ have this property, with $s_i = (X_i, Y_i)$ and $q_j = (x_j, y_j)$. For $\varepsilon > 0$, replace s_i by the block

$$Q_i = \big\{(X_i + \varepsilon^2 x_j, Y_i + \varepsilon y_j) : 1 \le j \le n\big\}. \tag{2.1}$$

For all sufficiently small ε, denote the resulting point set by $S[Q]$.

Lemma 2.1 (orientation rules)**.** *For sufficiently small positive ε, the following hold in $S[Q]$.*

1. *A triple in one block has its orientation from Q.*
2. *A triple in three different blocks has its orientation from S.*
3. *If the first two points of an increasing triple are in one block, its orientation is $-$.*
4. *If the last two points are in one block, its orientation is $+$.*

The parameter ε may be chosen rational, and $S[Q]$ again has both coordinates strictly increasing. If S and Q have rational coordinates, then so does $S[Q]$.

Proof. The first assertion follows by scaling within a block, and the second by continuity from the corresponding triple of cluster centres. If $p < q$ are in one block and z is in a later block, then

$$q - p = (\varepsilon^2 \Delta x, \varepsilon \Delta y), \qquad \Delta x, \Delta y > 0.$$

In the determinant $\det(q-p, z-p)$, the term $-\varepsilon\Delta y\,(z_x - p_x)$ dominates, proving the third assertion. The fourth is the reflected calculation.

Only finitely many strict coordinate inequalities and nonzero determinant conditions are involved. Hence they hold throughout some interval $0 < \varepsilon < \varepsilon_0$; choose a rational value there. Shrinking ε_0 if necessary also preserves the coordinate order. In an iteration, this choice is made afresh at every finite stage; no uniform perturbation parameter is asserted or needed. $\square$

For a point set R, let $c_j(R), u_j(R), v_j(R)$ denote the numbers of j-point caps, cups, and convex subsets. Put

$$C(R) = \sum_{j\ge1} c_j(R), \qquad U(R) = \sum_{j\ge1} u_j(R), \qquad W(R) = \sum_{j\ge1} v_j(R). \tag{2.2}$$

The same endpoint-cluster phenomenon is used by Baek and Balko [BB26, Lemma 14] to control the largest convex subset of an almost-vertical blow-up. In the present uniform setting, retaining all choices gives the following exact identities.

Lemma 2.2 (exact composition count)**.** *If $|S| = r$ and $|Q| = n$, then*

$$C(S[Q]) = C(Q) \sum_{j\ge1} c_j(S) n^{j-1}, \tag{2.3}$$

$$U(S[Q]) = U(Q) \sum_{j\ge1} u_j(S) n^{j-1}, \tag{2.4}$$

$$W(S[Q]) = rW(Q) + C(Q)U(Q) \sum_{j\ge2} v_j(S) n^{j-2}. \tag{2.5}$$

Proof. Consider a cap meeting at least two blocks. Only its first occupied block may contain more than one point: two points in any later block, together with a point in an earlier block, form a positive triple by Lemma 2.1. The first block contributes an arbitrary nonempty cap of Q, every other occupied block contributes one arbitrary point, and the occupied block indices form a cap of S. The orientation rules also show these conditions are sufficient. Summing over macro-caps proves (2.3); reflection proves (2.4).

Now let $X \subseteq S[Q]$ be convex and meet at least two blocks. Its upper and lower hull chains are respectively a cap and a cup with common endpoints. The lower hull contains at most one point from the first occupied block: two such points followed by any selected point in a later block would have sign $+$ along the lower chain, contradicting Lemma 2.1. Since every selected point is a hull vertex, all remaining points of the first block lie on the upper chain. Thus the first occupied block intersects X in a cap. The reflected argument makes the last-block intersection a cup. We claim every intermediate block contains at most one point.

Indeed, suppose selected points $a < b_1 < b_2 < c$ lie in that order, with b_1, b_2 in one intermediate block and a, c in earlier and later blocks. For small ε, the points b_1, b_2 lie on the same side of the line ac: this is true of their common cluster centre, and all relevant determinants are strict. If they lie above ac, convexity would put both on the upper chain from a to c, forcing $\chi(a, b_1, b_2) = -$; if they lie below, both lie on the lower chain, forcing $\chi(b_1, b_2, c) = +$. Both conclusions contradict Lemma 2.1.

Choose one representative from each occupied endpoint block and the unique point in every occupied intermediate block. This is a subset of the convex set X, hence is convex, and its order type is the corresponding subset of S. Thus the occupied block indices form a convex subset of S.

Conversely, fix a convex j-subset of S, $j \geq 2$. Choose a nonempty cap of Q in its first block, a nonempty cup of Q in its last block, and one point in each intermediate block. Along the upper macro-hull, insert the entire first-block cap and the rightmost point of the last-block cup; the orientation rules make the result a strict cap. The reflected construction along the lower macro-hull is a strict cup containing the remaining chosen points. These two chains share their endpoints, lie on opposite sides of their endpoint chord, and do not cross. Their union is therefore the boundary of a convex polygon. The choices are unique and contribute $C(Q)U(Q)n^{j-2}$. Sets contained in a single block contribute $rW(Q)$, which proves (2.5). □

3 Iteration of a fixed template

Fix a template S of size $r \geq 2$. Let its largest cap and cup have sizes a and b, and define

$$Q_0 = \{\text{one point}\}, \qquad Q_d = S[Q_{d-1}]. \tag{3.1}$$

Thus $|Q_d| = r^d$.

Proposition 3.1. *The iterates satisfy*

$$\lim_{d\to\infty} \frac{\log W(Q_d)}{(\log |Q_d|)^2} = \frac{a+b-2}{2\log r}. \tag{3.2}$$

Proof. Define the fixed polynomials

$$A(z) = \sum_{j\geq 1} c_j(S) z^{j-1}, \quad B(z) = \sum_{j\geq 1} u_j(S) z^{j-1}, \quad D(z) = \sum_{j\geq 2} v_j(S) z^{j-2}. \tag{3.3}$$

Their degrees are $a-1$, $b-1$, and at most $r-2$, respectively; each has positive leading coefficient, and D is nonzero because every pair is convex. From Lemma 2.2,

$$C(Q_d) = C(Q_{d-1})A(r^{d-1}), \qquad U(Q_d) = U(Q_{d-1})B(r^{d-1}). \tag{3.4}$$

Since a fixed positive polynomial P of degree e satisfies $\log P(r^t) = et\log r + O_S(1)$ uniformly for $t \geq 0$, summing (3.4) over the levels gives

$$\log C(Q_d) = (a-1)\binom{d}{2}\log r + O_S(d), \tag{3.5}$$

$$\log U(Q_d) = (b-1)\binom{d}{2}\log r + O_S(d). \tag{3.6}$$

The third recurrence unrolls exactly as

$$W(Q_d) = r^d W(Q_0) + \sum_{s=1}^{d} r^{d-s} C(Q_{s-1})U(Q_{s-1})D(r^{s-1}). \tag{3.7}$$

By (3.5)–(3.6), the logarithm of the sth summand is

$$(a+b-2)\binom{s-1}{2}\log r + O_S(s) + O_S(d-s). \tag{3.8}$$

There are only d summands, and the last one has the displayed quadratic term with $s = d$. More explicitly, since $D(r^{d-1}) \geq v_2(S) > 0$, the last summand gives

$$W(Q_d) \geq v_2(S)C(Q_{d-1})U(Q_{d-1}).$$

Hence upper-bounding the sum by d times its largest quadratic-scale bound and using this explicit lower bound yields

$$\log W(Q_d) = \frac{a+b-2}{2}d^2\log r + O_S(d). \tag{3.9}$$

Division by $(d\log r)^2$ proves the proposition. $\square$

4 Balanced cup–cap templates

We first record the geometric operation used in the classical recursive cup–cap construction. Write $A \prec B$ when every point of A precedes every point of B in x-order, internal orientations are preserved, and every increasing triple meeting both sides has sign $-$ if its first two points lie in A, and sign $+$ if its last two points lie in B.

Lemma 4.1 (strong separation)**.** *Given rational point sets A and B in general position, they have a rational realization $A \prec B$ in general position. Moreover, a cap in $A \prec B$ meeting both sides consists of a cap of A and one point of B, while a cup meeting both sides consists of one point of A and a cup of B.*

Proof. First shear both sets so that their two coordinates increase. Place a copy of A in the block $(\varepsilon^2 x, \varepsilon y)$ near $(0,0)$ and a copy of B in the block $(1+\varepsilon^2 x, 1+\varepsilon y)$ near $(1,1)$. The determinant calculation from Lemma 2.1 gives the two required mixed signs for all sufficiently small $\varepsilon > 0$. A rational ε may be selected below all finitely many strict thresholds. The cap and cup classifications follow immediately: two points of the later block forbid a cap, while two points of the earlier block forbid a cup; the mixed signs also prove the converse. $\square$

Now let $T_{m,0}$ and $T_{m,m}$ be singletons, and for $0 < i < m$ form

$$T_{m,i} = T_{m-1,i-1} \prec T_{m-1,i}. \tag{4.1}$$

using Lemma 4.1. These are the standard cup–cap examples. Up to the reflection $\rho(x, y) = (-x, y)$, they are the classical Erdős–Szekeres sets $P(i+2, m-i+2)$ in the notation of Baek and Balko [BB26, Section 4].

Lemma 4.2. *The cell $T_{m,i}$ has $\binom{m}{i}$ points, largest cap of size $i+1$, and largest cup of size $m-i+1$.*

Proof. The size recurrence is Pascal's identity. For caps, the rule in (4.1) gives the larger of the left-child cap size plus one and the right-child cap size. Induction gives $i + 1$ in either case. Reflection gives the cup statement. The boundary convention is consistent because the boundary cells are singletons. □

For $k \geq 3$, take the central cell

$$S_k = T_{2k-4,k-2}. \tag{4.2}$$

Then

$$r_k = |S_k| = \binom{2k-4}{k-2}, \qquad a_k = b_k = k - 1. \tag{4.3}$$

Proof of Theorem 1.1. Fix k and iterate S_k. By Proposition 3.1, configurations of size r_k^d satisfy

$$\frac{\log W(Q_d)}{(\log |Q_d|)^2} \longrightarrow \frac{k-2}{\log \binom{2k-4}{k-2}}. \tag{4.4}$$

For arbitrary N, take the least d with $r_k^d \geq N$ and retain any N points. Deletion creates no new convex subsets, while $r_k^d < r_k N$, so the normalized coefficient is unchanged. Consequently

$$\limsup_{N\to\infty} \frac{\log f(N)}{(\log N)^2} \leq \frac{k-2}{\log \binom{2k-4}{k-2}}. \tag{4.5}$$

The right-hand side is strictly larger than $1/2$ for every finite k, since $\binom{2k-4}{k-2} < 2^{2k-4}$, and tends to $1/2$ from above. Stirling's formula gives $\log \binom{2k-4}{k-2} = 2k - O(\log k)$. Letting $k \to \infty$ proves the theorem. □

The same calculation, together with the cup–cap theorem of Erdős and Szekeres [ES35], identifies the limitation of fixed-template iteration.

Proposition 4.3. *For every fixed template S, the coefficient in Proposition 3.1 is at least $1/2$.*

Proof. If the largest cap and cup have sizes a and b, the cup–cap theorem applied with forbidden sizes $a + 1$ and $b + 1$ gives

$$r \leq \binom{a+b-2}{a-1} \leq 2^{a+b-2}.$$

Thus $(a + b - 2)/(2 \log r) \geq 1/2$. □

5 Optimality for decomposable point sets

Following Balko, Kynčl, Langerman, and Pilz [BKLP17], a point set is *decomposable* if it is a singleton, or if it has a partition into decomposable sets L and R such that L lies to the left of and deep below R. Here deep below means that every point of L lies below every line through two points of R, while every point of R lies above every line through two points of L.

We use the mirrored convention. If L lies to the left of and deep below R, then for $\rho(x, y) = (-x, y)$ one has $\rho(R) \prec \rho(L)$. Conversely, reflecting a split $A \prec B$ gives such a deep-below split. The reflection ρ preserves cap and cup sizes and the number of convex subsets. We therefore call a point set *mirror-decomposable* if it is a singleton, or if it has a realization $A \prec B$ in which A and B are mirror-decomposable. These are precisely the ρ-images of decomposable sets. Equivalently, they are represented by ordered full binary trees whose internal nodes are the operation in Lemma 4.1.

Baek and Balko's theorem controls the largest convex subset in this class [BB26, Theorem 7]; the following theorem instead determines the asymptotic minimum of the total convex-subset count. Let

$$g(N) = \min\{W(P) : |P| = N,\ P \text{ mirror-decomposable}\}.$$

Theorem 5.1. *Every N-point mirror-decomposable set P satisfies*

$$\log W(P) \geq \frac{1}{2}(\log N)^2 - O((\log N)^{3/2}), \tag{1}$$

with an absolute implicit constant. Consequently

$$\lim_{N\to\infty} \frac{\log g(N)}{(\log N)^2} = \frac{1}{2}. \tag{2}$$

We need two preparatory estimates. At a node $T = A \prec B$, write $a = |A|$ and $b = |B|$. The strong-separation classification gives

$$C(T) = C(B) + (b+1)C(A), \tag{3}$$

$$U(T) = U(A) + (a+1)U(B). \tag{4}$$

Lemma 5.2 (cap–cup product)**.** *Every s-point mirror-decomposable set S satisfies*

$$\log C(S) + \log U(S) \geq \frac{1}{2}(\log s)^2 - \log s.$$

Proof. Put $R(S) = \sqrt{C(S)U(S)}$. The Cauchy–Schwarz inequality $\sqrt{(p+r)(q+t)} \geq \sqrt{pq} + \sqrt{rt}$ and (3)–(4) give

$$R(A \prec B) \geq \sqrt{b+1}\, R(A) + \sqrt{a+1}\, R(B).$$

Starting at S, follow a larger child until reaching a leaf. If the current subtree has m_i points, the followed child has $m_{i+1} = m_i - s_i$ points, where $1 \leq s_i \leq m_i/2$. Iterating the last display gives

$$\log R(S) \geq \frac{1}{2}\sum_i \log(s_i + 1). \tag{5}$$

Let $t_i = \log m_i$ and $d_i = t_i - t_{i+1}$. Since $0 < d_i \leq 1$, convexity of x^d on $0 \leq d \leq 1$, with $x = m_i/2$, shows

$$(m_i/2)^{d_i} \leq 1 + d_i(m_i/2 - 1) \leq s_i + 1;$$

here we used $d_i = -\log(1 - s_i/m_i) \le 2s_i/m_i$. Consequently

$$\sum_i \log(s_i + 1) \ge \sum_i d_i(t_i - 1) \ge \frac{1}{2}(\log s)^2 - \log s,$$

because $\sum_i d_i = \log s$ and $\sum_i d_i t_i = ((\log s)^2 + \sum_i d_i^2)/2$. Combining this with (5) gives $\log R(S) \ge \frac{1}{4}(\log s)^2 - \frac{1}{2}\log s$; doubling proves the claim. $\square$

We next refine the count by endpoints. For $p < q$, let $c(p, q)$ and $u(p, q)$ be the numbers of caps and cups with endpoints p, q. Let $x(p)$ count caps with left endpoint p, including the singleton, and let $y(q)$ count cups with right endpoint q, again including the singleton. Set

$$X = \max_p x(p), \qquad Y = \max_q y(q), \qquad M = \max\bigl(1, \max_{p<q} c(p, q)u(p, q)\bigr).$$

For a leaf, $(X, Y, M) = (1, 1, 1)$. Directly from strong separation,

$$X(A \prec B) = \max\{(b + 1)X(A), X(B)\}, \tag{6}$$
$$Y(A \prec B) = \max\{Y(A), (a + 1)Y(B)\}, \tag{7}$$
$$M(A \prec B) = \max\{M(A), M(B), X(A)Y(B)\}. \tag{8}$$

Indeed, for crossing endpoints $p \in A$, $q \in B$, one has exactly $c(p, q) = x_A(p)$ and $u(p, q) = y_B(q)$.

Every nonsingleton convex set has a unique upper cap and lower cup with the same endpoints, and conversely every such pair forms a convex set. Hence

$$W(S) = |S| + \sum_{p<q} c(p, q)u(p, q), \qquad M(S) \le W(S) \le 2|S|^2 M(S). \tag{9}$$

Also $x(p) = 1 + \sum_{q>p} c(p, q)$, and every two-point cup is available, so

$$X(S) \le |S|M(S), \qquad Y(S) \le |S|M(S). \tag{10}$$

Finally, $C(S) \le |S|X(S)$ and $U(S) \le |S|Y(S)$. Thus Lemma 5.2 implies, writing lower-case letters for base-two logarithms,

$$x(S) + y(S) \ge \frac{1}{2}(\log |S|)^2 - 3 \log |S|. \tag{11}$$

Proof of Theorem 5.1. It is enough to consider all sufficiently large N; increase the absolute constant to absorb the remaining values. Put $L = \log N$, $R = \lceil \sqrt{L} \rceil$, and $\lambda = \log L$. In the decomposition tree of P, start at the root and repeatedly follow a larger child. If n_i is the current size and s_i the discarded sibling size, stop when first

$$n_i < N/2^{4R}.$$

Call a level large when $s_i \ge n_i/L^2$.

Suppose first that fewer than R levels are large. A nonlarge level loses at most $2/L^2$ bits of size, since $-\log(1 - s_i/n_i) \le 2/L^2$, while every level loses at most one bit. The total loss before stopping exceeds $4R$, so there are more than $\frac{3}{2}RL^2$ nonlarge levels. At least half have their discarded sibling on the same side of the path. Fix one leaf in each such sibling and one leaf below the stopping node. Choosing an arbitrary subset of the fixed sibling leaves, together with the terminal leaf, gives distinct caps if the siblings lie on the right, and distinct cups if they lie on the left. This follows inductively upward from the mixed signs in Lemma 4.1. Therefore

$$\log W(P) > \frac{3}{4} R L^2,$$

which is stronger than (1).

Now suppose there are at least R large levels. At each of R selected large nodes, both children have at least $2^{L-\Delta}$ leaves, where

$$\Delta = 4R + 2\lambda + 1.$$

Indeed, the followed child has at least $n_i/2$, the sibling at least n_i/L^2, and $n_i \geq N/2^{4R}$ before the stop. Since $L - \Delta > 3$ for all sufficiently large L and $u \mapsto u^2/2 - 3u$ is then increasing in the relevant range, (11) gives uniformly for each such child

$$x + y \geq F := \frac{1}{2}(L - \Delta)^2 - 3L.$$

Let $\mu = \log M(P)$. Equations (6)–(8) make X, Y, M nondecreasing from a child to its parent. By (10), every coordinate of every subtree is at most $\mu + L$. If $\mu \geq F - L$, the desired estimate follows immediately. Otherwise put

$$D = F - \mu > L, \qquad \ell = D - L > 0.$$

Every coordinate of every child at a selected node is at least ℓ, by (11) and the ceiling $\mu + L$.

Order the R selected nodes from deepest to highest. At the deepest node $A \prec B$, the crossing term in (8) gives $x_A + y_B \leq \mu$. Together with the two radial bounds (11), this implies

$$x_B, y_A \geq 2D - L.$$

Thus both coordinates of the parent are at least $h_0 := 2D - L$.

At a later selected node, let H be the child containing the preceding selected node and Q the opposite child. If H is the left child, then $x_H + y_Q \leq \mu$ and $x_Q + y_Q \geq F$, whence

$$x_Q \geq x_H + D.$$

The parent increases its x-coordinate by at least D and preserves its y-coordinate. If H is the right child, the reflected argument increases y by at least D and preserves x. Intermediate nodes cannot erase either gain, by monotonicity.

Among the $R-1$ later attachments, one direction occurs q_* times, where $q_* \geq \lceil (R-1)/2 \rceil$. At its final occurrence the forward product uses a path coordinate of at least $h_0 + (q_* - 1)D$ and an opposite-child coordinate of at least ℓ. Therefore

$$\mu \geq (q_* + 2)D - 2L, \qquad \mu \geq \frac{q_* + 2}{q_* + 3} F - \frac{2L}{q_* + 3}.$$

Since $\Delta = O(\sqrt{L})$, $q_* = \Omega(\sqrt{L})$, and $F = \frac{1}{2}L^2 - O(L^{3/2})$, both alternatives give

$$\mu \geq \frac{1}{2} L^2 - O(L^{3/2}).$$

Now $W(P) \geq M(P)$, proving (1).

For the limit in (2), observe that every Q_d constructed from the Pascal templates is mirror-decomposable. Indeed, at every internal split of the macro strong tree, triples whose two same-side points lie in one micro-block have the required sign by Lemma 2.1(3)–(4), while triples using distinct micro-blocks inherit the sign of the macro split. Induction therefore identifies the iterate with the strong tree obtained by substituting the micro strong tree at every macro leaf. An induced subset is also strong after suppressing empty and unary nodes. The arbitrary-N deletion argument in the proof of Theorem 1.1 therefore gives $\limsup \log g(N)/(\log N)^2 \le 1/2$, while the first part gives the matching liminf. □

6 Verification artifact

The accompanying exact-arithmetic program independently checks the finite combinatorics behind Lemma 2.2. It constructs the abstract composition signs, realizes them with rational coordinates, counts caps and cups by a last-edge dynamic program, and counts convex subsets through upper/lower endpoint chains. A 9-point composition is additionally checked by enumerating all 2^9 subsets. On $S = Q = T_{4,2}$, the substitution formulas and independent dynamic programs both return

$$(C, U, W) = (14136, 14136, 441399)$$

for the resulting 36-point composition. The artifact is described in the supplementary README and is not used as a premise of the proof. A second exact-arithmetic program checks the one-reset inequalities used in Theorem 5.1 over finite boxes of logarithmic states. It is an algebraic smoke test, not a substitute for the multiscale argument.

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